

SMOLENSK, Russian Federation

WMO: 267810

Lat: 54.7491N

Lon: 32.0647E

Elev: 781

StdP: 14.29

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.6	-1.8	-12.3	2.9	-6.7	-6.8	3.9	-0.5	18.7	30.0	16.4	27.0	3.4	90	0.542

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	82.6	68.2	79.5	66.4	76.5	65.0	69.7	79.4	68.0	77.2	66.3	74.5	5.5	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.3	99.7	74.7	64.6	93.9	72.5	63.1	88.9	70.7	34.2	79.5	32.7	77.3	31.4	74.3	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
15.1	13.2	11.6	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-14.6	87.2	5.3	3.9	-18.5	90.0	-21.6	92.3	-24.5	94.5	-28.4	97.4
			-15.0	72.1	5.2	1.9	-18.7	73.5	-21.8	74.6	-24.7	75.6	-28.6	77.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	42.6	20.4	21.4	30.2	44.3	54.6	60.8	65.4	62.1	52.5	42.2	32.1	24.1	(6)
(7)		DBStd	17.95	10.85	11.30	8.14	8.68	8.29	6.27	5.61	6.12	6.50	7.24	8.59	10.68	(7)
(8)		HDD50	4241	917	802	615	216	47	2	0	0	47	256	536	803	(8)
(9)		HDD65	8330	1382	1222	1080	623	334	157	64	131	377	706	986	1268	(9)
(10)		CDD50	1551	0	0	0	44	190	325	477	377	122	16	0	0	(10)
(11)		CDD65	162	0	0	0	0	12	29	77	42	2	0	0	0	(11)
(12)		CDH74	1249	0	0	0	5	110	197	572	344	21	0	0	0	(12)
(13)		CDH80	250	0	0	0	0	11	19	127	92	1	0	0	0	(13)
(14)	Wind	WSAvg	5.4	6.3	6.1	6.1	5.6	5.2	4.8	4.2	4.2	4.3	5.3	6.1	6.2	(14)
(15)	Precipitation	PrecAvg	26.5	1.7	1.3	1.6	1.7	2.0	2.8	3.7	2.6	2.5	2.3	2.3	2.2	(15)
(16)		PrecMax	34.5	3.4	3.4	3.4	3.1	3.6	4.8	7.5	7.2	8.9	5.5	4.3	3.5	(16)
(17)		PrecMin	20.0	0.3	0.2	0.2	0.3	0.6	0.5	0.2	0.7	0.5	0.4	1.1	0.8	(17)
(18)		PrecStd	3.6	0.9	0.7	0.8	0.8	0.8	1.2	1.9	1.4	1.8	1.2	0.9	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.5	40.5	55.9	73.3	81.3	82.5	87.9	88.9	77.0	65.9	52.1	46.3	(19)
MCWB			39.1	38.0	45.6	55.4	65.7	67.5	70.3	68.8	64.3	56.2	50.1	44.8	(20)	
2%		DB	37.0	37.5	49.0	67.8	77.3	79.3	83.4	81.5	70.9	58.2	47.8	41.8	(21)	
		MCWB	35.5	35.2	40.4	52.6	62.6	66.6	68.7	67.5	60.3	53.7	46.2	40.7	(22)	
5%		DB	34.9	35.6	44.4	62.9	72.6	76.0	80.1	77.4	66.7	55.2	45.4	38.4	(23)	
		MCWB	33.8	34.0	38.3	50.8	59.9	64.7	67.6	65.3	58.2	51.8	43.9	37.4	(24)	
10%		DB	33.1	34.0	40.6	58.6	68.7	72.8	76.7	73.5	63.0	52.4	43.0	35.6	(25)	
		MCWB	32.3	32.8	36.3	48.4	57.7	63.0	65.8	63.4	56.2	49.6	41.8	34.7	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.7	38.7	46.1	57.0	67.9	70.0	72.8	71.0	65.5	57.8	50.7	45.2	(27)
MCD B			40.7	39.8	53.9	69.6	79.4	78.7	83.6	82.3	73.2	61.2	52.0	45.9	(28)	
2%		WB	35.9	35.8	41.8	53.9	64.3	67.8	70.3	68.8	61.8	54.8	46.7	40.9	(29)	
		MCD B	36.6	36.7	47.1	64.3	74.3	76.4	80.2	79.1	68.6	57.7	47.5	41.6	(30)	
5%		WB	33.9	34.2	39.1	51.8	61.8	66.1	68.5	66.9	59.1	52.2	44.1	37.7	(31)	
		MCD B	34.6	35.1	42.8	61.1	70.0	74.1	77.6	75.4	64.5	54.6	45.1	38.4	(32)	
10%		WB	32.4	33.0	36.8	49.7	59.3	64.1	66.9	64.8	57.0	50.1	41.9	34.9	(33)	
		MCD B	33.0	33.9	40.2	57.5	66.6	71.0	75.2	71.6	61.8	52.0	42.8	35.5	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.2	9.9	11.9	15.7	17.3	16.1	16.7	16.9	15.5	11.1	7.4	7.2	(35)
MCD BR			6.2	7.6	16.5	21.6	22.9	20.3	21.3	22.5	21.5	15.5	9.8	6.7	(36)	
MCWBR			6.4	6.9	10.1	10.5	10.7	9.8	9.4	10.1	11.6	10.3	8.4	6.3	(37)	
5% WB		MCD BR	6.2	7.0	14.1	19.7	19.7	18.3	19.4	20.0	18.5	13.2	8.7	6.7	(38)	
		MCWBR	6.4	6.7	9.5	9.9	10.4	9.9	9.6	9.8	10.9	9.9	8.0	6.5	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.286	0.320	0.345	0.407	0.391	0.372	0.400	0.414	0.369	0.346	0.325	0.288	(40)	
(41)		taud	2.145	2.091	2.143	2.170	2.309	2.379	2.317	2.272	2.371	2.392	2.256	2.161	(41)	
(42)		Ebn at Noon	200	233	258	258	271	278	266	253	251	228	185	171	(42)	
(43)		Edh at Noon	20	30	36	41	38	36	37	37	29	23	18	16	(43)	
(44)	All-Sky Solar Radiation	RadAvg	209	492	949	1268	1670	1788	1726	1391	922	438	162	123	(44)	
(45)		RadStd	46	79	121	140	131	165	159	157	86	89	32	36	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+1.17	N/A	N/A	N/A	N/A	N/A	N/A	-378	N/A	N/A
(47)	Regional (8 neighbors)	+1.12	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page